

## Embedded SQL (Self-Learning)

Q. What is Embedded SQL Discuss how SQL statements interact with host programming languages using Host Variables and Cursors. Differentiate between Static and Dynamic SQL.

### [DEF] Definition to Remember

Embedded SQL is a technique where SQL statements are inserted directly into the source code of a general-purpose host programming language (C, C++, Java, COBOL). It bridges the gap between application logic and database interaction. A pre-compiler processes the embedded SQL before the host language compiler.

### 1. Why Embedded SQL

SQL alone lacks general-purpose constructs (UI, file I/O, loops, networking). Embedding SQL in a host language allows full-featured applications to interact with databases.

### 2. Key Mechanisms

#### A. Host Variables

Variable

#### B. Cursors

A stand

|                 |                                                   |
|-----------------|---------------------------------------------------|
| DECLARE cursor  | -> Define the cursor + SELECT query               |
| OPEN cursor     | -> Execute the query                              |
| FETCH (in loop) | -> Retrieve one row at a time into host variables |
| CLOSE cursor    | -> Release resources                              |

### 3. Static SQL vs Dynamic SQL

| Feature                      | Static SQL                                    | Dynamic SQL                                   |
|------------------------------|-----------------------------------------------|-----------------------------------------------|
| <b>**SQL text known at**</b> | <b>**Compile time**</b> (hard-coded)          | <b>**Runtime**</b> (built as strings)         |
| <b>**Pre-compilation**</b>   | Yes -- validated and optimized before         | No -- compiled at execution time              |
| <b>**Performance**</b>       | <b>**Faster**</b> (pre-optimized plan)        | <b>**Slower**</b> (compiled on-the-fly)       |
| <b>**Security**</b>          | <b>**Very Secure**</b> (immune to SQL Injecti | <b>**Risky**</b> (vulnerable to SQL Injection |
| <b>**Flexibility**</b>       | Low -- cannot adapt SQL at runtime            | <b>**High**</b> -- can build any query dynami |
| <b>**Use Case**</b>          | Standard reports, fixed operations            | User-driven search screens, flexible q        |

### [MUST-WRITE] Must-Write Points (for 10 marks)

1. Embedded SQL = SQL statements inserted directly into a host language (C, Java, COBOL).
2. A pre-compiler processes embedded SQL before the host language compiler runs.
3. Host Variables (prefixed with :) bridge data between the app and SQL.
4. Cursors handle multi-row result sets row by row: DECLARE -> OPEN -> FETCH -> CLOSE.
5. Static SQL: hard-coded at compile time; pre-optimized, secure, fast, inflexible.
6. Dynamic SQL: built at runtime; flexible, but slower and vulnerable to SQL Injection.
7. Static SQL is preferred for security; Dynamic SQL is used when query structure must change.

### [RECALL] Quick Recall

Embedded SQL -> Host Language + SQL -> Host Variables (:prefix, bridges data) -> Cursors (row-by-row: DECLARE->OPEN->FETCH->CLOSE) -> Static (compile-time, fast, secure) -> Dynamic (runtime, flexible, injection risk)